

## Education

### 2023–2027 Sharif University of Technology, Tehran, Iran

B.Sc., Electrical Engineering (Digital Systems focus) • expected Aug 2027

- **Cumulative GPA: 16.88/20.** Above the department average (16.49) and university average (16.09) for this cohort, as printed on the Sharif transcript.
- Most recent semester (Spring 2026): **19.30/20** over 19 units.
- **National rank 134** among about 150,000 candidates in the Mathematics and Physics group of the Iranian University Entrance Exam (Konkur).
- **Relevant coursework (grade/20):** Fundamentals of Programming (20.0); Object-Oriented Programming (17.8); Foundations of Data Science (19.3); Embedded Systems (19.8); Computer Interfacing, a graduate course (19.0); Computer Structure and Microprocessor with lab (17.0). In progress (Fall 2026): Data Structures and Algorithms; Parallel Programming.

## Research Interests

- Data platforms, analytics and recommender systems
- Software engineering and developer automation
- Embedded and real-time software
- Machine learning systems and hardware acceleration
- Statistical and probabilistic methods for data-driven systems

## Projects

### 2026–present SUT Music: Data Platform and Recommendation Engine

[source](#)

Foundations of Data Science course, Dr. Salavati • Python, scikit-learn, PostgreSQL, FastAPI, React

- **Data:** Led the scraping and cleaning of a Telegram music community's history into a database of 14,800+ tracks and 86,900+ reactions.
- **Recommenders:** Built two generations (matrix factorization, then a two-stage artist-to-track ensemble) behind a backend API and a Telegram Web App.
- **Engineering:** Set up automated deployment of the web app and of each recommender engine version to the server on every push.
- **Team:** Team of 3; project lead.

### 2024–present EE Department Bot Ecosystem and BotOS Framework

Lead developer • Python (asyncio), PostgreSQL, Docker, Google Sheets API

- **System:** Built and maintain bots for course-file distribution, assignment grading and re-judge ticketing, used by the EE department and by two student groups.
- **Framework:** Designed BotOS, which deploys new bots from declarative JSON configuration in isolated Docker containers.
- **Adoption:** Began with the programming courses I assisted (Fundamentals of Programming, Object-Oriented Programming); the platform has since been adopted by a second university (IUST).

### Spring 2026 Smart Surveillance System on an Embedded Linux Board

[source](#)

Embedded Systems course, Dr. Gholampour • Orange Pi Zero Plus 2 (H5), C, Python

- **System:** Individual project: an on-board person-detection system in which a Haar-cascade detector feeds an HTTPS dashboard, a REST API and an MQTT/e-mail alert daemon.
- **Reliability:** Added a software watchdog that restarts the detector when the camera feed stalls, hysteresis-based thermal throttling and an event log.
- **Evaluation:** Measured boot time, detection accuracy under varied lighting, concurrent load and temperature with and without a fan; documented in a technical report.

- Spring 2026 **OTP Memory Controller (RTL design, synthesis and verification)** [source](#)  
*ASIC/FPGA System Design course, Dr. Baharvand • Verilog, Cadence Genus and Xcelium*
- **Design:** Co-designed a power-on FSM controller that reads a 256-bit OTP array, applies hardware patches and checks a CRC-16.
  - **Team and flow:** Team of 4, led week 1 (rotating lead); verified with a testbench, then ran synthesis and gate-level simulation.
- Spring 2025 **MIPS32 Processor Implementation on FPGA**  
*Computer Architecture lab, Dr. Movahedin • Verilog, MIPS32 assembly, Xilinx Zynq-7010*
- **Design and validation:** Single-cycle, multi-cycle and pipelined MIPS32 CPUs; the multi-cycle one ran on an FPGA board.

## Teaching Experience

- 2024–2026 **Teaching Assistant, Fundamentals of Programming and OOP**  
*Sharif University of Technology • Dr. Vosoughi Vahdat and Dr. Arasteh • four consecutive semesters*
- Fall 2025 **Teaching Assistant, Electronics 1**  
*Sharif University of Technology • Dr. Fakharzadeh, Dr. Sarvari, Dr. Medi, Dr. Besharatirad*
- Fall 2025 **Lab Assistant, Logic Circuit Lab (FPGA)**  
*Sharif University of Technology • Dr. Movahedin • supported students designing on Zynq-7010 FPGA boards*
- Spring 2026 **Teaching Assistant, Theory of Circuits**  
*Iran University of Science and Technology (IUST)*

## Industrial Training

- Summer 2026 **Trainee, Parsian Zarin (ABLY): MQL5 expert advisors; Excellent Pass**

## Technical Skills

- **Languages:** Python, C, C++, Java, SQL, Verilog, MIPS32 assembly.
- **Software engineering:** backend and REST API development, asynchronous programming, relational databases (PostgreSQL, SQLite), Docker, Git, CI/CD, Linux, React (web front end).
- **Data and machine learning:** data analysis, machine learning, recommender systems.
- **Embedded and hardware:** embedded Linux on single-board computers, FPGA design with Xilinx Vivado, Cadence Genus and Xcelium.

## Languages

Persian (native); English (official test score pending).